Week 3

Monday, September 2, 2024 12:56 PM

AQ 3.1

Q1. _ _ _ _ _ _ _ _ _ _ _

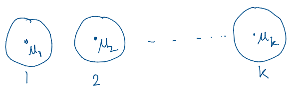
$(10)^{100}$ — A

Q2. C , B

Q3.

$$\mu_k = \frac{\sum_{i=1}^{n} x_i \mathbb{1}(z_i = k)}{\sum_{i=1}^{n} \mathbb{1}(z_i = k)} \quad \text{— (E)}$$

Q4.



$\min \| x_i - \mu_{z_i} \|^2$

$x_1 - \mu$

Cluster

x_1 — z_1

x_2 — z_2

$\vdots$ — $\in \{1, \dots, k\}$

x_n — z_n

x_1, x_2, x_3, x_4, x_5

$z_i \in \{1, \dots, k\}$

$(x_2 - \mu_{z_2})^2 + (x_4 - \mu_{z_2})^2$

$(x_2 - \mu_1)^2 + (x$

$(x_1 - \mu_{z_1})^2 + (x_2 - \mu_{z_2})^2 + \dots + (x_n - \mu_{z_n})^2$

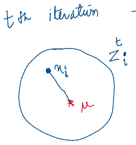
AQ 3.2

Q1. (A)

Q2. B

Q3. B , ~~D~~ C

Q4. t th iteration — B



AQ 3.3

Q1. D

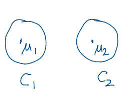
Q2. B

Q3. ~~B~~ A

Q4. A

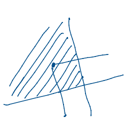
AQ 3.4

Q1.



— (C)

Q2.



— (A)

Q3. (B) , (C)

Q4. (B)

AQ 3.5

$\rightarrow X_1, X_2, \dots, X_n$

(X_1, z_1)

(X_2, z_2)

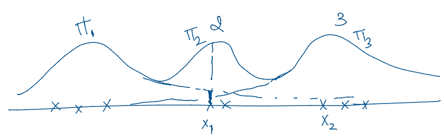
$\vdots$

(X_n, z_n)

$z_i \in \{1, 2, \dots, k\}$

$(X_i \mid z_i = l) \sim N($

$P(z_i = l) = \pi_l$



$(X_1 \mid z_1 = 2) \sim N(\mu_{z_1}, \sigma_{z_1}^2)$

$(X_2 \mid z_2 = 3) \sim N(\mu_{z_2}, \sigma_{z_2}^2)$

$P(z_1 = 1) = \pi_1$

$P(z_1 = 2) = \pi_2$

$P(z_1 = 3) = \pi_3$

$P(z_2 = 1) = \pi_1$

$P(x_i \text{ coming from cluster } j) = \pi_j \quad , \forall i: 1, 2, \dots, n \quad \forall j: 1, 2, \dots, k$

$X_1 : \{\lambda_1^1, \lambda_2^1, \lambda_3^1\}$

$P(X_1 \in \text{cluster } 1) = \lambda_1^1$

$P(X_1 \in \text{cluster } 2) = \lambda_2^1$

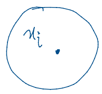
$P(X_1 \in \text{cluster } 3) = \lambda_3^1$

$\rightarrow X_i : \{\lambda_1^i, \lambda_2^i, \lambda_3^i, \dots, \lambda_k^i\}$

Q1. ~~C~~ A

Q2. ~~A~~ C

Q3. x_1 x_2 x_3 x_4 $\boxed{x_5}$



Q4. (B)

AQ 3.6

Q1 B

Q2. C

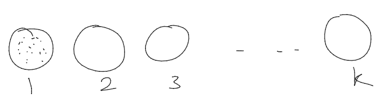
Q3. B

Q4. C, D

Week - 3 [Clustering] — Unsupervised.

Dataset : $\{x_1, x_2, \ldots, x_n\} \rightarrow x_i \in \mathbb{R}^d$

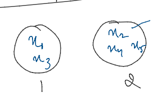
\# clusters = k



z_i: 1 2 1 2 2

$\{x_1, x_2, x_3, x_4, x_5\}$

k = 2



Each datapoint has 2 possibilities $= \boxed{2^5}$

2 2 _ . 2

$\rightarrow \{x_1, x_2, \ldots x_n\}$ K clusters

\# Partitions $= k^n$

Goodness of this partition

$\{x_1, x_2, \ldots x_n\}$, k clusters

$\downarrow \quad \downarrow \quad \quad \downarrow$

$z_1 \quad z_2 \ldots \quad z_n \rightarrow$ cluster indicators

$z_1, z_2, \ldots, z_n \in \{1, 2, \ldots k\}$

Define $F(z_1, z_2, \ldots z_n) = \sum_{i=1}^{n} \| x_i - \mu_{z_i} \|_2^2$

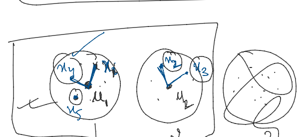
μ_{z_i}: average of z_i^{th} cluster.

Goal : $\min F(z_1, \ldots, z_n)$

$x = (x_1, x_2)$

$\|x\|_2^2 = x_1^2 + x_2^2$

$\boxed{\|x\|_1 = \sum_{i=1}^{2} |x_i| \; : L_1 \text{ norm}}$



$z_i = k$

$\searrow$ average of z_i^{th} cluster.

Goal: $\min F(z_1, \dots, z_n)$

$$\mu_k = \frac{\sum_{i=1}^{n} x_i \, \mathbb{1}(z_i = k)}{\sum_{i=1}^{n} \mathbb{1}(z_i = k)} \quad , \quad \mathbb{1}(z_i = k) = \begin{cases} 1, & z_i = k \\ 0, & \text{o.w.} \end{cases}$$

$$\mu_1 = \frac{\sum_{i=1}^{n} x_i \mathbb{1}(z_i = 1)}{\sum_{i=1}^{n} \mathbb{1}(z_i = 1)} = \frac{x_1 \underbrace{\mathbb{1}(z_1=1)}_{1} + \cancel{x_2 \underbrace{\mathbb{1}(z_2=1)}_{0}} + \cancel{x_3 \underbrace{\mathbb{1}(z_3=1)}_{0}} + x_4 \underbrace{\mathbb{1}(z_4=1)}_{1} + x_5 \underbrace{\mathbb{1}(z_5=1)}_{1}}{\underbrace{\mathbb{1}(z_1=1)}_{1} + \underbrace{\mathbb{1}(z_2=1)}_{0} + \underbrace{\mathbb{1}(z_3=1)}_{0} + \underbrace{\mathbb{1}(z_4=1)}_{1} + \underbrace{\mathbb{1}(z_5=1)}_{1}}$$

$z_2 = 2$

$$\boxed{\mu_1 = \frac{x_1 + x_4 + x_5}{3}}$$

Goal: To minimize $F(z_1, z_2, \dots, z_n) = \sum_{i=1}^{n} \| x_i - \mu_{z_i} \|_2^2$

$\boxed{k^{n}}$

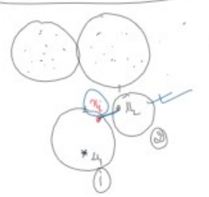
Algorithm: Lloyd's algorithm / k-means clustering.

$\rightarrow$ $z_1^0, z_2^0, z_3^0, \dots, z_n^0 \in \{1, 2, \dots, k\}$

Initialization

① Randomly choose k data pts. uniformly

② k-means ++



Step 1: Calculate the means for every k

At any iteration t, $\mu_k^t = \dfrac{\sum_{i=1}^{n} x_i \, \mathbb{1}(z_i^t = k)}{\sum_{i=1}^{n} \mathbb{1}(z_i^t = k)}$

Step 2: Re-assignment

For every data pt., compute $z_i^{t+1} = \arg\min_k \| x_i - \mu_k^t \|_2^2$

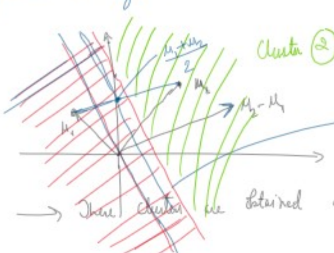
At (t+1)th iteration,

Lloyd's algorithm converges!

Pf.

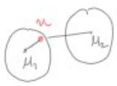
Voronoi region

μ_1, μ_2



$$x^T(\mu_2 - \mu_1) = \frac{\|\mu_2\|^2 - \|\mu_1\|^2}{2}$$

$$x = \frac{\mu_1 + \mu_2}{2}$$

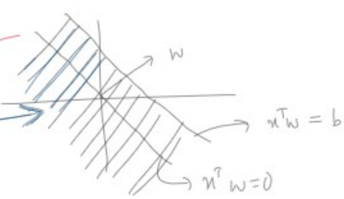


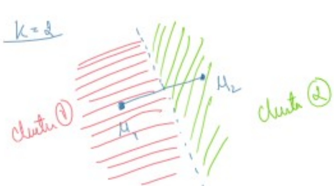
$\longrightarrow$ These clusters are obtained after Lloyd's algo.

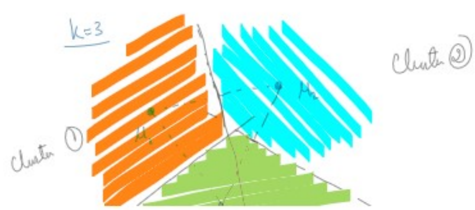
$\| x - \mu_1 \|_2^2 \le \| x - \mu_2 \|^2$ [From Lloyd's algo.]

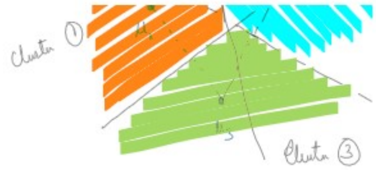
$$\boxed{x^T(\mu_2 - \mu_1) \le \frac{\|\mu_2\|^2 - \|\mu_1\|^2}{2}}$$

$$\boxed{x^T w \le b}$$









Initialization

① From the set of data points, we choose n datapts uniformly at random.

② k-means ++ (Probabilistic way).

$$\mu_1^0, \underbrace{\mu_2^0, \ldots, \mu_k^0}$$

$\rightarrow \mu_1^0 \rightarrow \{ \text{———} \}$

μ_2^0

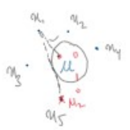
$$\text{Score}(x) = \min_{j=1,2,\ldots,l-1} \left(\| x - \mu_j^0 \|^2 \right), \ \forall x$$

$$\underline{\mu_2^0}: \quad \text{Score}(x_i) = \min_{j=1} \left(\| x_i - \mu_1^0 \|_2^2 \right) = d(x_i, \mu_1^0)^2 = s_i$$

$$\boxed{\mu_2^0 \sim \left(\frac{s_1}{\sum_{i=1}^n s_i}, \frac{s_2}{\sum_{i=1}^n s_i}, \ldots, \frac{s_n}{\sum_{i=1}^n s_i} \right)}$$

$$\text{Score}(x_i) = \min_{j=1,2,\ldots,l-1} \left(\| x - \mu_j^0 \|^2 \right), \ \forall x$$

$$\text{Score}(x) = \min_{j=1,2,\ldots,l-1} \left(\| x - \mu_j^0 \|^2 \right), \ \forall x$$



$\mu_2^0 = x_5 = \frac{25}{75}$

$\underline{\mu_2^0}$

$$\text{Score}(x_1) = \min_{j=1} \left(\| x_1 - \mu_1^0 \|^2 \right) = \| x_1 - \mu_1^0 \|_2^2 = d(x_1, \mu_1^0)^2 = a_1 = \underset{(\text{Say})}{5}$$

$$\text{Score}(x_2) = \| x_2 - \mu_1^0 \|^2 = d(x_2, \mu_1^0)^2 = a_2 = 10$$

$$a_3 = 15$$

$$a_4 = 20$$

$$\text{Score}(x_5) = \| x_5 - \mu_1^0 \|^2 = d(x_5, \mu_1^0)^2 = a_5 = 25$$

$$\mu_2^0 \sim \left(\frac{a_1}{\sum_{i=1}^5 a_i}, \frac{a_2}{\sum_{i=1}^5 a_i}, \ldots, \frac{a_5}{\sum_{i=1}^5 a_i} \right), \quad \mu_2^0 \sim \left(\frac{5}{75}, \frac{10}{75}, \frac{15}{75}, \frac{20}{75}, \frac{25}{75} \right)$$

$$\text{Score}(x) = \min_{j=1,\ldots,l-1} \left(\| x - \mu_j^0 \|^2 \right)$$

$$\underline{\mu_1^0, \mu_2^0}, \underbrace{\mu_3^0, \mu_4^0, \ldots, \mu_k^0}$$

$\underline{\mu_3^0}$

$$\text{Score}(x_1) = \min_{j=1,2} \left(\| x_1 - \mu_1^0 \|^2, \| x_1 - \mu_2^0 \|^2 \right) = b_1$$

$$\text{Score}(x_2) = \min \left(\| x_2 - \mu_1^0 \|^2, \| x_2 - \mu_2^0 \|^2 \right) = b_2$$

$$\vdots$$

$$\text{Score}(x_5) = \min \left(\| x_5 - \mu_1^0 \|^2, \| x_5 - \mu_2^0 \|^2 \right) = b_5$$

$$\mu_3^0 \sim \left(\frac{b_1}{\sum b_i}, \frac{b_2}{\sum b_i}, \ldots, \frac{b_5}{\sum b_i} \right)$$

Q How do you choose k ?

$$F(z_1, z_2, \ldots, z_n) = \underbrace{\sum_{i=1}^n \| x_i - \mu_{z_i} \|^2}$$

Look for (F + (Penalty)(P))

Clustering

$\left.\begin{matrix} \{x_1, x_2, \ldots, x_n\} \\ z_1, z_2, \ldots, z_n \end{matrix}\right\} \text{— } K$ clusters

→ k^n partitions are possible

→ Lloyd's algorithm (?)

$F(z_1, z_2, \ldots, z_n) = \sum_{i=1} \| x_i - \mu_{z_i} \|$

Look for $F + \text{Penalty}(P)$

$\uparrow$ ing f^n

The least value of $(F + P)$

k = # clusters

$\boxed{k = 2}$

$k = 3$

$\rightarrow$ k^n partitions are possible

$\rightarrow$ Lloyd's algorithm (?)

$z_1^0, z_2^0, \ldots, z_n^0$

μ_k^t

z_i^{t+1}

$\rightarrow$ Voronoi

$\rightarrow$ k

PDF

Solve with...

Course: Machine Learning Technique
Sep 2024

1. Given the following 2D dataset with 6 points:

Lloyd's algo.

$$A = (1,2), \quad B = (1,4),$$
$$C = (1,0), \quad D = (10,2),$$
$$E = (10,4), \quad F = (10,0).$$

Perform k-means clustering with $k = 2$:

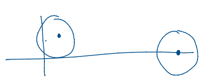
(a) Initialize centroids: $C_1 = (1,2)$, $C_2 = (10,2)$.

(b) Assign points to the nearest centroid based on Euclidean distance.

(c) List the final clusters and their centroids.

	$\|x_i - \mu_1^0\|^2$	$\|x_i - \mu_2^0\|^2$
A. $(1,2)$		
B: $(1,4)$		
$C = (1,0)$		
$D = (10,2)$		
$E = (10,4)$		
$F = (10,0)$		

C_1: $(1,0)$, $(1,2)$, $(1,4)$ $\qquad$ C_2: $(10,2)$, $(10,0)$, $(10,4)$



μ_1^1, μ_2^1

C_1: $(1,0)$, $(1,2)$, $(1,4)$ $\qquad$ C_2: $(10,2)$, $(10,0)$, $(10,4)$

$\mu_1^1 = (1,2)$ $\qquad$ $\mu_2^1 = (10,2)$

$\mu_1^1 = (1,2) = \mu_1^0$

$\mu_2^1 = (10,2) = \mu_2^0$

$\mu_1^1 = (2,2)$

$\mu_2^1 = (10,1)$

$(1,0), (1,2), (1,4) \rightarrow (1,3)$

$(10,2), (10,0), (10,4) \rightarrow (10,3$

Solutions:

Step 1: Initialization

We initialize the centroids as follows:

$$C_1 = (1,2), \quad C_2 = (10,2)$$

Step 2: Assignment of Points

Next, we calculate the Euclidean distance from each point to the centroids and assign each point to the nearest centroid.

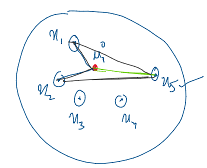
Distances to C_1 and C_2:

$d(A, C_1) = \sqrt{(1-1)^2 + (2-2)^2} = 0 \qquad d(A, C_2) = \sqrt{(1-10)^2 + (2-2)^2} = 9$

2 ways of initializations

① Randomly choose k clusters uniformly

② $\boxed{k\text{-means } ++}$, k - clusters



μ_1^0 : first cluster mean

This is randomly selected

$\mu_1^0, \mu_2^0, \ldots, \mu_k^0$

$\boxed{\mu_2^0}$

$\text{Score}(x) = \min_{j=1} \left(\| x - \mu_j^0 \|^2 \right) = \left(\| x - \mu_1^0 \|^2 \right)$

$\text{Score}(x_1) = \| x_1 - \mu_1^0 \|^2 = d_1 = 10$

$\text{Score}(x_2) = \quad = d_2 = 15$

$\vdots \quad = 5$

$\quad = 10$

$\text{Score}(x_5) = \quad = d_5 = 20$

x_i	x_1	x_2	x_3	x_4	x_5
$P(x_i)$	$\frac{10}{60}$	$\frac{15}{60}$	$\frac{5}{60}$	$\frac{10}{60}$	$\frac{20}{60}$

$\mu_2^0 \sim \left(\frac{d_1}{\sum_{i=1} d_i}, \frac{d_2}{\sum_{i=1} d_i}, \ldots, \frac{d_5}{\sum_{i=1} d_i} \right)$: $P(\mu_2^0 = x_1) = \frac{d_1}{\sum d_i}$

$P(\mu_2^0 = x_2) = \frac{d_2}{\sum_{i=1}^{5} d_i}$

$\mu_1^0 \quad \mu_2^0$

μ_1^0, μ_2^0

$\mu_3^0 =$

$\text{score}(x) = \min_{j=1,2} \left(\| x - \mu_j^0 \|^2 \right)$

$= \min \left\{ \| x - \mu_1^0 \|^2, \| x - \mu_2^0 \|^2 \right\}$

$\text{Score}(x_1) = \min \left\{ \| x_1 - \mu_1^0 \|^2, \| x_1 - \mu_2^0 \|^2 \right\}$

$= a_1 \rightarrow$

a_2

a_3

a_4

$(x_5) \quad a_5$

Solutions:

Step 1: Initialization

We initialize the centroids as follows:

$$C_1 = (1, 2), \quad C_2 = (10, 2)$$

Step 2: Assignment of Points

Next, we calculate the Euclidean distance from each point to the centroids and assign each point to the nearest centroid.

Distances to C_1 and C_2:

$$\begin{aligned}
d(A, C_1) &= \sqrt{(1-1)^2 + (2-2)^2} = 0 & d(A, C_2) &= \sqrt{(1-10)^2 + (2-2)^2} = 9 \\
d(B, C_1) &= \sqrt{(1-1)^2 + (4-2)^2} = 2 & d(B, C_2) &= \sqrt{(1-10)^2 + (4-2)^2} = 9.22 \\
d(C, C_1) &= \sqrt{(1-1)^2 + (0-2)^2} = 2 & d(C, C_2) &= \sqrt{(1-10)^2 + (0-2)^2} = 9.22 \\
d(D, C_1) &= \sqrt{(10-1)^2 + (2-2)^2} = 9 & d(D, C_2) &= \sqrt{(10-10)^2 + (2-2)^2} = 0 \\
d(E, C_1) &= \sqrt{(10-1)^2 + (4-2)^2} = 9.22 & d(E, C_2) &= \sqrt{(10-10)^2 + (4-2)^2} = 2 \\
d(F, C_1) &= \sqrt{(10-1)^2 + (0-2)^2} = 9.22 & d(F, C_2) &= \sqrt{(10-10)^2 + (0-2)^2} = 2
\end{aligned}$$

Assignments:

$$\begin{aligned}
A &\to C_1 \\
B &\to C_1 \\
C &\to C_1 \\
D &\to C_2 \\
E &\to C_2 \\
F &\to C_2
\end{aligned}$$

Step 3: Update Centroids

Now we calculate the new centroids based on the assigned points.

$$C_1 = \frac{A + B + C}{3} = \frac{(1, 2) + (1, 4) + (1, 0)}{3} = (1, 2)$$

$$C_2 = \frac{D + E + F}{3} = \frac{(10, 2) + (10, 4) + (10, 0)}{3} = (10, 2)$$

Step 4: Repeat Assignment

Since the centroids did not change, we will reassign the points again. The assignments remain the same.

Final Assignments:

$$\begin{aligned}
A &\to C_1 \\
B &\to C_1 \\
C &\to C_1 \\
D &\to C_2 \\
E &\to C_2 \\
F &\to C_2
\end{aligned}$$

Final Clusters

The final clusters and their centroids are:

- **Cluster 1:** Points A, B, C with centroid $C_1 = (1, 2)$
- **Cluster 2:** Points D, E, F with centroid $C_2 = (10, 2)$

$$\text{score}(x) = \min_{j=1,2} \left(\| x - \mu_j^0 \|^2 \right)$$

$$= \min \left\{ \underbrace{\| x - \mu_1^0 \|^2}, \underbrace{\| x - \mu_2^0 \|^2} \right\}$$

$$\text{Score}(x_1) = \min \left\{ \| x_1 - \mu_1^0 \|^2, \| x_1 - \mu_2^0 \|^2 \right\}$$

$$= a_1 \rightarrow$$

a_2

a_3

a_4

(x_5) a_5

Llyod's algo.

(1) Initialization: $z_1^0, z_2^0, \ldots, z_n^0 \in \{1, 2, \ldots, k\}$

Until convergence

Step (1): Compute mean for every k.

$$\mu_k^t = \frac{\sum_{i=1}^{n} x_i \mathbb{1}(z_i^t = k)}{\sum_{i=1}^{n} \mathbb{1}(z_i^t = k)}$$

Step (2): Reassignment:

$$\forall i, \quad z_i^{t+1} = \arg\min_k \| x_i - \mu_k^t \|^2$$

This algo. always converges!

2. With respect to the Lloyd's algorithm, choose the correct statements:
   - A. The partition configurations cannot repeat themselves.
   - B. After doing the reassignments (consider at least one point reassigned to the new cluster), we might get the same means for all clusters.
   - C. Objective function after making the re-assignments strictly reduces.
   - D. Objective function after making the re-assignments may increase.
   - E. A change in the objective function's value indicates that the partition configuration has changed.
   - F. For partitioning n data points across k partitions, Lloyd's algorithm takes k^n iterations to converge.

3. Which of the following are true about k-means and k-means++?

$\| x_i - \mu_j \|^2$

   - A. Choosing different cluster center can result in different clusters.
   - B. k-means cluster data by separating data points into groups based on distance from cluster centroids.
   - C. It may happen that the k-means algorithm does not converge.
   - D. Poor initialization can lead to slow convergence.
   - E. k-means++ initializes cluster centroids randomly, just like the standard k-means algorithm.
   - F. ~~Both~~ k-means ~~and k-means++~~ can guarantee finding the global optimum of the clustering problem.

4. A k-means++ algorithm with $k = 3$ was applied to the following 2D points:

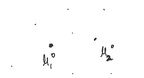
$$\{ (2, 2), (3, 3), (4, 4), (10, 10), (12, 12), (13, 13). \quad \mu_2^0$$

The point $(2, 2)$ is chosen as the first cluster mean. $\mu_1^0 = (2,2)$, $\mu_2^0 = ?$, $\mu_3^0 = ?$

(a) Which point has the highest probability of being chosen as the 2nd cluster mean? Use the Manhattan distance to compute the distances.

(b) If the point with the highest probability of being chosen as the 2nd cluster mean is indeed chosen, then which point has the highest probability of being chosen as the third cluster mean? Would it be certainly chosen as the third cluster mean?

(x_1, y_1), (x_2, y_2) $\longrightarrow |x_1 - x_2| + |y_1 - y_2|$



μ_2^0: Calculate scores for (x_i, y_i).

$\text{Score}((2,2)) = 0 \rightarrow 0$

$\text{Score}((3,3)) = |1| + |1| = 2 \rightarrow \frac{2}{64}$

$\text{Score}(4,4) = 4 \longrightarrow 4/64$

$\text{Score}(10,10) = 16 \longrightarrow 16/64$

$\text{Score}(12,12) = 20 \longrightarrow 20/64$

$\text{Score}(13,13) = 22 \longrightarrow 22/64$

$\mu_1^0 = (2,2)$, $\mu_2^0 = (13,13)$, $\mu_3^0 = ?$

Solutions:

Answer 1(a) Using the k-means++ initialization, the second cluster mean is chosen based on the distance from the first cluster mean $(2, 2)$. The algorithm selects a point probabilistically, where points farther from the first mean are more likely to be selected.

Using the Manhattan distance (also known as L1 norm), the distances from the first cluster mean $(2, 2)$ to each point are computed as follows:

$$\begin{aligned}
d((2,2),(3,3)) &= |3-2| + |3-2| = 2 \\
d((2,2),(4,4)) &= |4-2| + |4-2| = 4 \\
d((2,2),(10,10)) &= |10-2| + |10-2| = 16 \\
d((2,2),(12,12)) &= |12-2| + |12-2| = 20 \\
d((2,2),(13,13)) &= |13-2| + |13-2| = 22
\end{aligned}$$

Since k-means++ chooses the next cluster mean with a probability proportional to the distance from the already chosen mean, the point with the largest Manhattan distance from $(2, 2)$ will have the highest probability of being selected.

Thus, the point $(13, 13)$ has the highest probability of being chosen as the 2nd cluster mean.

Answer(b):

If $(13, 13)$ is chosen as the 2nd cluster mean, we now calculate the Manhattan distance from both cluster means, $(2, 2)$ and $(13, 13)$, to each remaining point. The next point to be chosen is based on its minimum distance from any of the two existing cluster means, as the k-means++ algorithm attempts to spread the cluster centers as much as possible.

The Manhattan distances to each point from both means are:

$$\begin{aligned}
d((2,2),(3,3)) &= |3-2| + |3-2| = 2, & d((13,13),(3,3)) &= |13-3| + |13-3| = 20 \\
d((2,2),(4,4)) &= |4-2| + |4-2| = 4, & d((13,13),(4,4)) &= |13-4| + |13-4| = 18 \\
d((2,2),(10,10)) &= |10-2| + |10-2| = 16, & d((13,13),(10,10)) &= |13-10| + |13-10| = 6 \\
d((2,2),(12,12)) &= |12-2| + |12-2| = 20, & d((13,13),(12,12)) &= |13-12| + |13-12| = 2
\end{aligned}$$

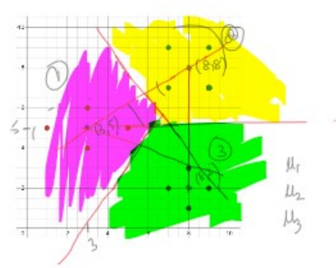
For each point, the minimum distance to either $(2, 2)$ or $(13, 13)$ is:

$$\begin{aligned}
\min d(3,3)) &= 2 \\
\min d(4,4) &= 4 \\
\min d((10,10) &= 6 \\
\min d((12,12) &= 2
\end{aligned}$$

Since $(10, 10)$ has the largest minimum distance of 6, it has the highest probability of being chosen as the 3rd cluster mean.

Would it certainly be chosen? Not necessarily. In k-means++, the selection is probabilistic, and while (10, 10) has the highest probability of being selected, it is not guaranteed to be the 3^{rd} cluster mean. Other points can still be chosen based on their respective probabilities.

5. We have [illegible] data points in $\mathbb{R}^2$. There are three clusters. Each colour in the image below represents a cluster.



(a) Find the center of each cluster.

(b) Draw the voronoi regions.

6. Consider the following 2D dataset:

$$\left[\ (1,1),(2,1),(4,3),(5,4),(6,6),(8,7).\ \right]$$

Let us select two initial cluster centroids: $\underset{\mu_1^0}{(2,1)}$ and $\underset{\mu_2^0}{(6,6)}$. Based on the above information answer the following questions:

After one iteration of k-means, let C_1 and C_2 denote the updated centroids.

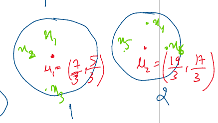
(a) What is the value of $C_1 + C_2$?

(b) What is the value of the objection function after one iteration?

(c) If we choose k-means++ with k = 3, then are with what probability the third cluster centroids will be chosen if the first two (2, 1) and (6, 6) based on k-means++ initialization.

$$F(z_1, \dots, z_6) = \sum_{i=1}^{6} \|x_i - \mu_{z_i}\|_2^2 \quad , \quad \mu_1^0 = (2,1), \ \mu_2^0 = (6,6)$$

x_i	$\|x_i - \mu_1^0\|^2$	$\|x_i - \mu_2^0\|^2$	z_i
(1, 1)	1	50	1
(2, 1)	0	41	1
(4, 3)	8 ✓	13	1
(5, 4)	18	5 ✓	2
(6, 6)	41	0	2
(8, 7)	72	5	2



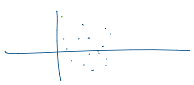
$$\mu_1^1 = \left(\frac{7}{3}, \frac{5}{3}\right) \quad , \quad \mu_2^1 = \left(\frac{19}{3}, \frac{17}{3}\right)$$

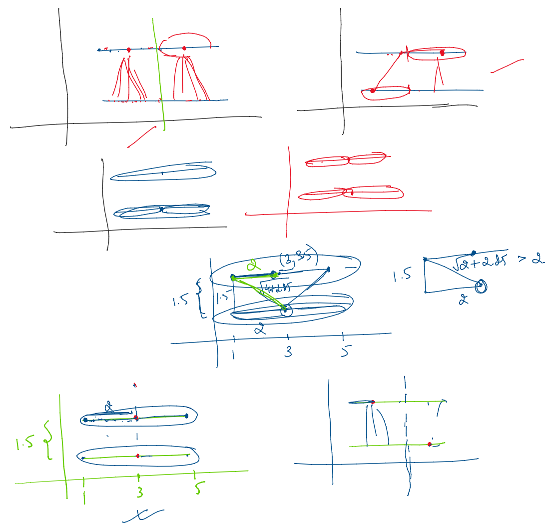
$$F(z_1, z_2, \dots, z_6) = \sum_{i=1}^{6} \|x_i - \mu_{z_i}\|_2^2$$

$$= \left\|x_1 - \left(\frac{7}{3}, \frac{5}{3}\right)\right\|_2^2 + \dots + \left\|x_3 - \left(\frac{7}{3}, \frac{5}{3}\right)\right\|_2^2$$

$$+ \left\|x_4 - \left(\frac{19}{3}, \frac{17}{3}\right)\right\|_2^2 + \dots + \dots \ x_5 \qquad x_6$$

k = 2





Solutions

Answer 2:

- **Correct statements:**
  - **The partition configurations cannot repeat themselves.**
    - *Reason:* In Lloyd's algorithm, once a partition configuration has been established, it cannot revert back to a previous state as the means update and the reassignment of points is dependent on the current means.
  - **Objective function after making the re-assignments strictly reduces.**
    - *Reason:* The k-means algorithm is designed to minimize the objective function (usually the sum of squared distances from points to their respective cluster centers) at each iteration, ensuring that the function does not increase after a reassignment.
  - **A change in the objective function's value indicates that the partition configuration has changed.**
    - *Reason:* If the objective function value changes, it suggests that either the means have changed or the assignments of points to clusters have altered, thus indicating a new partition configuration.
- **Incorrect statements:**
  - **After doing the reassignments, we might get the same means for all clusters.**
    - *Reason:* Although it's possible for means to remain unchanged, this is not guaranteed, as different points can shift between clusters leading to a change in means.
  - **Objective function after making the re-assignments may increase.**
    - *Reason:* The objective function is expected to decrease or stay the same after reassignments, but it should never increase as per the design of the algorithm.
  - **For partitioning n data points across k partitions, Lloyd's algorithm takes k^n iterations to converge.**
    - *Reason:* The complexity of Lloyd's algorithm is not k^n. Instead, the algorithm converges in a number of iterations that is dependent on the data distribution and cluster initialization, typically much lower than k^n.

Answer 3

The dataset consists of the following points:

$$A = (1, 2), \qquad B = (1, 4),$$
$$C = (1, 0), \qquad D = (10, 2),$$
$$E = (10, 4), \qquad F = (10, 0)$$